Talazoparib plus enzalutamide in men with first-line metastatic castration-resistant prostate cancer (TALAPRO-2): a randomised, placebo-controlled, phase 3 trial

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Summary

Background Co-inhibition of poly(ADP-ribose) polymerase (PARP) and androgen receptor activity might result in antitumour efficacy irrespective of alterations in DNA damage repair genes involved in homologous recombination repair (HRR). We aimed to compare the efficacy and safety of talazoparib (a PARP inhibitor) plus enzalutamide (an androgen receptor blocker) versus enzalutamide alone in patients with metastatic castration-resistant prostate cancer (mCRPC).

Methods TALAPRO-2 is a randomised, double-blind, phase 3 trial of talazoparib plus enzalutamide versus placebo plus enzalutamide as first-line therapy in men (age ≥18 years [≥20 years in Japan]) with asymptomatic or mildly symptomatic mCRPC receiving ongoing androgen deprivation therapy. Patients were enrolled from 223 hospitals, cancer centres, and medical centres in 26 countries in North America, Europe, Israel, South America, South Africa, and the Asia-Pacific region. Patients were prospectively assessed for HRR gene alterations in tumour tissue and randomly assigned (1:1) to talazoparib 0·5 mg or placebo, plus enzalutamide 160 mg, administered orally once daily. Randomisation was stratified by HRR gene alteration status (deficient *vs* non-deficient or unknown) and previous treatment with life-prolonging therapy (docetaxel or abiraterone, or both: yes *vs* no) in the castration-sensitive setting. The sponsor, patients, and investigators were masked to talazoparib or placebo, while enzalutamide was open-label. The primary endpoint was radiographic progression-free survival (rPFS) by blinded independent central review, evaluated in the intention-to-treat population. Safety was evaluated in all patients who received at least one dose of study drug. This study is registered with ClinicalTrials.gov (NCT03395197) and is ongoing.

Findings Between Jan 7, 2019, and Sept 17, 2020, 805 patients were enrolled and randomly assigned (402 to the talazoparib group and 403 to the placebo group). Median follow-up for rPFS was 24·9 months (IQR 21·9–30·2) for the talazoparib group and 24·6 months (14·4–30·2) for the placebo group. At the planned primary analysis, median rPFS was not reached (95% CI 27·5 months–not reached) for talazoparib plus enzalutamide and 21·9 months (16·6–25·1) for placebo plus enzalutamide (hazard ratio 0·63; 95% CI 0·51–0·78; p<0·0001). In the talazoparib group, the most common treatment-emergent adverse events were anaemia, neutropenia, and fatigue; the most common grade 3–4 event was anaemia (185 [46%] of 398 patients), which improved after dose reduction, and only 33 (8%) of 398 patients discontinued talazoparib due to anaemia. Treatment-related deaths occurred in no patients in the talazoparib group and two patients (<1%) in the placebo group.

Interpretation Talazoparib plus enzalutamide resulted in clinically meaningful and statistically significant improvement in rPFS versus standard of care enzalutamide as first-line treatment for patients with mCRPC. Final overall survival data and additional long-term safety follow-up will further clarify the clinical benefit of the treatment combination in patients with and without tumour HRR gene alterations.

Funding Pfizer.

Introduction

Prostate cancer is the second most common malignancy in men, with approximately 1·4 million new cases and 375 000 deaths in 2020 worldwide.[1] Despite recent approval of new agents and declining mortality in many high-income countries,[1,2] metastatic castration-resistant prostate cancer (mCRPC) remains lethal and the need for novel therapeutic strategies persists. Alterations in homologous recombination repair (HRR) genes, including *BRCA1/2* (hereafter referred to as *BRCA*), found in approximately one-quarter of patients with advanced prostate cancer, can sensitise patients to

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Research in context

Evidence before this study

In early 2017, we searched PubMed for relevant preclinical or clinical research published on DNA damage repair, homologous recombination repair (HRR), so-called BRCAness, novel hormonal therapies, androgen receptor signalling inhibition, poly(ADP-ribose) polymerase (PARP) inhibitors, and advanced prostate cancer. Despite recent approvals of new agents and declining mortality in many high-income countries, metastatic castration-resistant prostate cancer (mCRPC) is lethal and the need for novel therapeutic strategies remains. Current standard of care for first-line treatment of mCRPC comprises therapies targeting androgen receptor signalling together with androgen deprivation therapy. Co-inhibition of the androgen receptor signalling pathway and PARP might result in enhanced benefit with the potential to extend efficacy of PARP inhibitors to tumours regardless of HRR gene alterations. Two recent randomised phase 3 trials, PROpel and MAGNITUDE, investigated the combination of a PARP inhibitor together with the androgen biosynthesis inhibitor abiraterone as first-line treatment for mCRPC. The PROpel trial showed improved radiographic progression-free survival (rPFS) with olaparib plus abiraterone versus placebo plus abiraterone irrespective of HRR gene alteration status. The MAGNITUDE trial showed improved rPFS in patients with HRR gene alterations, including the *BRCA1/2* subgroup, but not in patients without HRR gene alterations.

Added value of this study

We present the primary results from the TALAPRO-2 study, the first phase 3 study evaluating the novel combination of talazoparib and enzalutamide versus a standard of care (enzalutamide) in patients with mCRPC. The study prospectively assessed the HRR gene alteration status in tumour tissue, considered a gold standard for establishing the biomarker status in cancer. Furthermore, the study used HRR status (HRR positive *vs* HRR negative or unknown status) as a prespecified stratification factor to establish benefits in each group. Lastly, we report an exploratory analysis showing the benefit of the combination of talazoparib plus enzalutamide specifically in patients who were established to have HRR gene alteration-negative mCRPC by prospective tumour tissue testing. In this rigorously conducted trial, the novel combination of talazoparib plus enzalutamide improved outcomes compared with placebo plus enzalutamide in patients with mCRPC with and without tumour HRR gene alterations.

Implications of all the available evidence

The recent randomised phase 3 trials, PROpel and MAGNITUDE, reported differing results in patients with mCRPC without HRR gene alterations. Although PROpel described a benefit of olaparib plus abiraterone for patients with or without HRR gene alterations, these alterations were retrospectively assessed, and randomisation was not stratified by HRR gene alteration status. MAGNITUDE showed a benefit of niraparib plus abiraterone only for patients with HRR gene alterations including the *BRCA1/2* subgroup. The reasons for these differing results are yet to be understood but might be related to the trial design or differences in the dosing or characteristics of the individual PARP inhibitors. The results of the TALAPRO-2 trial help to better elucidate the biological and clinical benefit of PARP inhibitor plus novel hormonal therapy combinations in prostate cancer with or without HRR gene alterations. Future research will focus on unravelling the molecular mechanisms underlying the efficacy of PARP inhibition and androgen receptor inhibitor combinations, especially in those patients whose prostate cancers did not harbour HRR gene alterations.

treatment with poly(ADP-ribose) polymerase (PARP) inhibitors.[3–7]

Enzalutamide, a current standard of care approved to treat CRPC and metastatic castration-sensitive prostate cancer, competitively inhibits androgen binding to the androgen receptor.[8] Talazoparib is a potent PARP inhibitor and traps PARP on single-strand DNA breaks, preventing DNA repair.[4,9] In the phase 2 TALAPRO-1 trial, talazoparib monotherapy (1 mg/day) showed durable antitumour activity and a manageable safety profile in patients with heavily pretreated mCRPC with HRR deficiency.[10] The phase 3 PROfound (olaparib) and TRITON3 (rucaparib) trials also showed benefit using a PARP inhibitor as monotherapy for patients with pretreated mCRPC who had HRR alterations, with prolonged progression-free survival (PFS) versus control treatment.[11,12]

Co-inhibition of the androgen receptor and PARP might result in enhanced benefit with the potential to extend efficacy of PARP inhibitors to tumours regardless of HRR gene alterations.[13] The underlying biology includes activation of PARP and the downregulation of HRR gene expression (inducing so-called BRCAness) by androgen receptor blockade,[14,15] and the role of PARP in supporting androgen receptor activity.[16,17]

The phase 3 TALAPRO-2 trial is assessing talazoparib combined with enzalutamide as first-line treatment in patients with mCRPC in two cohorts with equally split alpha for data analysis: unselected (cohort 1, the all-comers cohort, recruited first) and selected (cohort 2, HRR-deficient only, which completed recruitment after completion of enrolment in cohort 1) for DNA damage response alterations in genes directly or indirectly involved in HRR.[18] Here, we report the efficacy and safety results from the all-comers cohort at the time of the final analysis of the primary endpoint.

Methods

Study design and participants

TALAPRO-2 is an ongoing double-blind, randomised, placebo-controlled trial that enrolled patients from

223 hospitals, cancer centres, and medical centres in 26 countries in North America, Europe, Israel, South America, South Africa, and the Asia-Pacific region. The trial is being conducted in accordance with the International Ethical Guidelines for Biomedical Research Involving Human Subjects, Good Clinical Practice guidelines, the principles of the Declaration of Helsinki, and local laws. The protocol and amendments were approved by the institutional review board and independent ethics committee for each site. Details of the trial design have been published,[18] and are available in the protocol (appendix pp 87–94); important protocol deviations were recorded.

Eligible patients were adult men (age ≥18 years [≥20 years in Japan]) who were receiving ongoing androgen deprivation therapy, had asymptomatic or mildly symptomatic mCRPC, had an Eastern Cooperative Oncology Group (ECOG) performance status score of 0 or 1, had progressive disease at study entry (prostate-specific antigen [PSA] only or imaging-based), had adequate bone marrow function (including haemoglobin ≥9 g/dL), and had not received previous life-prolonging systemic therapy for CRPC or mCRPC.[18] Previous docetaxel and abiraterone or orteronel in the castration-sensitive setting were allowed. A full list of criteria is in the appendix (pp 13–15). All participants provided written informed consent.

Randomisation and masking

From an initial non-randomised open-label run-in, the starting dose of talazoparib in combination with 160 mg daily enzalutamide was determined to be 0·5 mg daily, based on drug–drug interaction effects with enzalutamide and pharmacokinetics demonstrating similar talazoparib exposure levels to 1 mg monotherapy daily.[18,19] Further details are available in the appendix (p 10).

Subsequent patients were randomly assigned 1:1 by site personnel, using a centralised interactive web response system and a permuted block size of 4, to talazoparib plus enzalutamide or matching placebo plus enzalutamide. Randomisation was stratified by previous novel hormonal therapy or docetaxel for castration-sensitive prostate cancer (yes *vs* no) and HRR gene alteration status (deficient *vs* non-deficient or unknown). The sponsor, patients, and investigators were masked to talazoparib or placebo, while enzalutamide was open-label.

Enrolment began in the all-comers population (including both patients with and without HRR gene alterations), cohort 1. Once enrolment was complete in cohort 1, enrolment continued but was restricted to patients with HRR gene alterations (cohort 2).

Procedures

Before randomisation, patients consented to provide solid tumour tissue (de novo or archival) or blood-based samples, or both, which were prospectively assessed at a central laboratory for the alteration status of DNA damage response genes directly or indirectly involved in HRR (*BRCA1*, *BRCA2*, *PALB2*, *ATM*, *ATR*, *CHEK2*, *FANCA*, *RAD51C*, *NBN*, *MLH1*, *MRE11A*, *CDK12*) using FoundationOne CDx or FoundationOne Liquid CDx (Foundation Medicine, Cambridge, MA, USA). A protocol amendment (Feb 26, 2020) allowed concurrent prospective circulating tumour DNA (ctDNA) testing in addition to the tumour tissue testing. Previous (ie, historical) testing results of tumour tissue performed using FoundationOne CDx could also be used. HRR gene alteration status was considered unknown if the sample was insufficient or inadequate, or if not meeting specified quality control metrics led to a test failure.

See Online for appendix

While the randomisation stratification combined non-deficient and unknown status, we also extracted HRR gene alteration status derived only from prospective tumour tissue testing. This enabled analysis of outcomes in a specific HRR-non-deficient subpopulation.

Patients in the talazoparib group received talazoparib 0·5 mg orally once daily as two 0·25 mg capsules, with or without food (moderate renal impairment, 0·35 mg daily as one 0·25 mg capsule and one 0·1 mg capsule) plus enzalutamide 160 mg orally once daily as four 40 mg capsules administered at the same time as talazoparib. Patients in the placebo group received placebo as two capsules administered orally with or without food plus enzalutamide matched to that received by the talazoparib group. Study treatment continued until radiographic progression by blinded independent central review, an adverse event leading to permanent discontinuation, patient decision to discontinue treatment, or death. After radiographic progression, treatment could be continued if the investigator determined benefit was still being derived.

Imaging scans (CT of chest, CT or MRI of abdomen and pelvis, and whole-body bone scan) were conducted every 8 weeks through to week 25 and every 12 weeks thereafter. Soft tissue responses were confirmed by a follow-up radiographic assessment at least 4 weeks later with no evidence of confirmed bone disease progression on repeated bone scans at least 6 weeks apart per Prostate Cancer Working Group 3 criteria. Clinical laboratory evaluations and safety assessments were done at screening and at scheduled visits, every 2 weeks up to week 17, every 4 weeks up to week 53, every 8 weeks thereafter while on study drug, then 28 days after discontinuation of all study treatments or before initiation of a new antineoplastic or investigational therapy, whichever occurred first. PSA assessments were carried out every 8 weeks until determination of radiographic progression. PSA progression or response was confirmed by a second consecutive value at least 21 days later. PSA was also assessed at week 53 and every 8 weeks thereafter. Circulating tumour cell enumeration was performed at screening, weeks 1, 9, 17, and 25, and the safety follow-up visit using the CELLSEARCH assay

(Menarini Silicon Biosystems, Bologna, Italy) run centrally at Covance/LabCorp (Indianapolis, IN, USA; Geneva, Switzerland; Shanghai, China)

Outcomes

The primary endpoint was radiographic PFS (rPFS), assessed by blinded independent central review per Response Evaluation Criteria in Solid Tumours (RECIST version 1.1; soft tissue disease) and Prostate Cancer Clinical Trials Working Group 3 (bone disease).[18]

Secondary endpoints were overall survival, rPFS (investigator-assessed), objective response rate (in patients with measurable soft tissue disease at baseline per RECIST 1.1) and duration of soft tissue response (both by blinded independent central review), proportion of patients with PSA response of 50% or greater, time to PSA progression, time to initiation of cytotoxic chemotherapy, time to initiation of subsequent antineoplastic therapy, time to first symptomatic skeletal event, time to disease progression or death on the first subsequent antineoplastic therapy for prostate cancer (investigator-assessed), time to opiate use for prostate cancer pain (to be reported separately), safety (incidence of adverse events characterised by type, severity [graded by National Cancer Institute Common Terminology Criteria for Adverse Events[20] version 4.03], timing, seriousness, and relationship to study treatment), pharmacokinetics (to be reported separately), and patient-reported outcomes (to be reported separately).

Exploratory subgroup analyses for rPFS by blinded independent central review were performed by HRR gene alteration status (classed as deficient *vs* non-deficient or unknown by randomisation stratification and classed as deficient, non-deficient, or unknown by prospective tumour tissue testing), and by baseline characteristics (age [<70 years *vs* ≥70 years], geographical region [Asia, EU and UK, North America, or rest of the world], ECOG performance status [0 *vs* 1], Gleason score [<8 *vs* ≥8], stage [M0 *vs* M1], type of progression [PSA only *vs* imaging-based with or without PSA progression], baseline PSA [<median *vs* ≥median], site of metastasis [bone only *vs* soft tissue only *vs* bone and soft tissue], and previous taxane or novel hormonal therapy for castration-sensitive prostate cancer [yes *vs* no]). Post-hoc analysis of rPFS (blinded independent central review) by *BRCA* status (*BRCA1/2*-altered, non-*BRCA1/2*-altered, and non-*BRCA1/2*-altered or unknown) was also performed.

Statistical analysis

Efficacy was analysed in the intention-to-treat population. Approximately 750 patients were to be enrolled in the

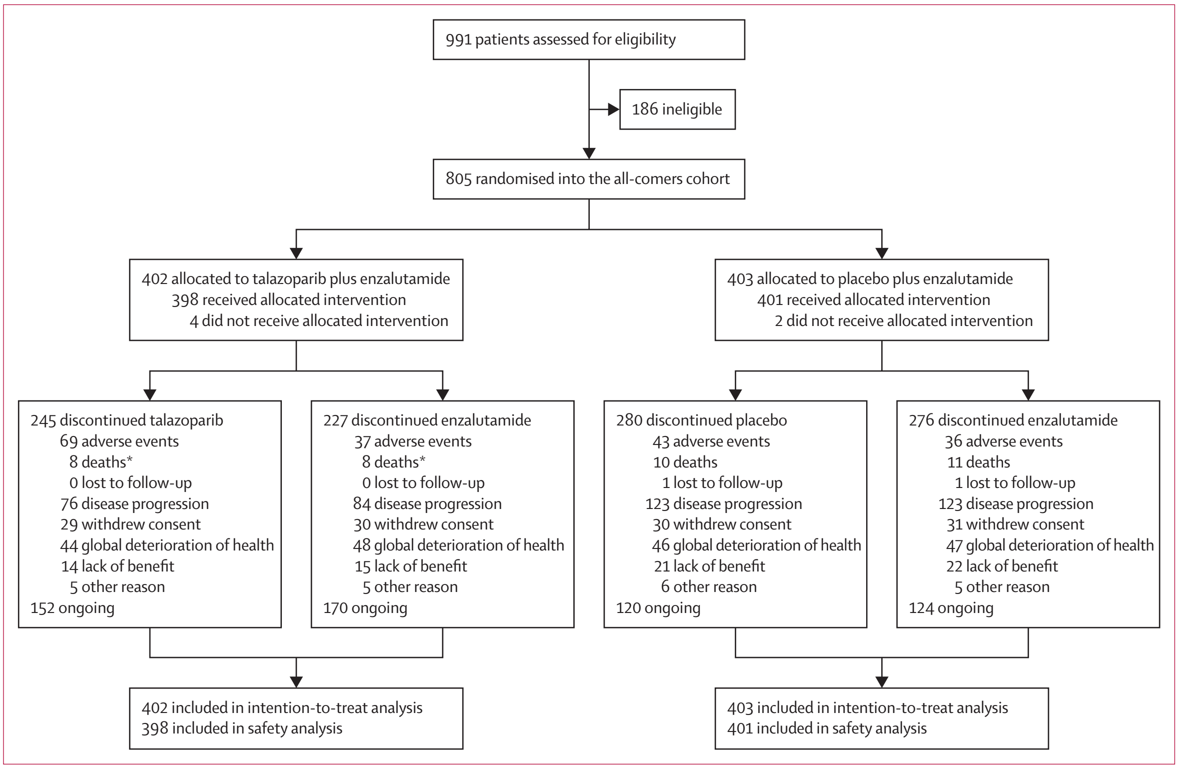


Figure 1: **Trial profile**

Two patients, one in each treatment group, who received enzalutamide but not talazoparib/placebo were excluded from the list of patients who discontinued treatment. One patient withdrew from the study with no protocol deviation reported; the second patient had a protocol deviation reported and discontinued from the study due to no longer meeting eligibility criteria. Important protocol deviations reported in ≥10% of patients are included in appendix p 16. *Discontinuation due to death was reported for one patient in error. The patient discontinued both talazoparib and enzalutamide due to the same adverse event on the same date.

all-comers cohort. To maintain overall type I error, alpha was split equally between the all-comers cohort and the ongoing HRR selected population. A one-sided test was used to show superiority of the active treatment (talazoparib over placebo). The overall one-sided alpha for the study was controlled at 0·025, which was equivalent to two-sided 0·05.

Sample size and power calculations were based on a stratified log-rank test and were performed using the software package East 6 (Cytel, Cambridge, MA, USA). For the primary comparison in the all-comers population, 333 rPFS events would provide 85% power to detect a hazard ratio (HR) of 0·696 using a one-sided stratified log-rank test at a significance level of 0·0125 based on a two-look group sequential design. A prespecified interim futility analysis of rPFS was planned using the O'Brien-Fleming β-spending function when approximately 50% of the expected events (167 rPFS events) had occurred. The interim rPFS results were assessed by an external data monitoring committee, and the all-comers cohort would be stopped for futility if the boundary was crossed at the interim analysis.

Overall survival was tested only if the rPFS results showed statistically significant improvement in a hierarchical stepwise procedure to preserve the overall type I error. For overall survival in the all-comers population, 438 overall survival events would provide 77% power to detect an HR of 0·75 using a one-sided stratified log-rank test at a significance level of 0·0125 based on a three-look group sequential design with O'Brien-Fleming α-spending function. There were two interim analyses of overall survival in the all-comers population: the first at the time of the final analysis of rPFS and the second when approximately 75% of the total required events (327 overall survival events) had occurred. Other endpoints had no adjustment for multiplicity.

Time-to-event endpoints were compared between treatment arms using a stratified log-rank test unless otherwise stated. HRs and associated two-sided 95% CIs were estimated by a Cox proportional hazards model. Median time-to-event endpoints were estimated by the Kaplan-Meier method, and 95% CIs were based on the Brookmeyer-Crowley method. For the subgroup analysis of rPFS (except for by *BRCA* status) and as shown in the forest plots, the HR was based on an unstratified Cox model with treatment as the only covariate due to the small number of patients in some of the subgroups. Missing or partial dates were imputed as specified per protocol. Other missing data were not imputed. SAS version 9.4 statistical software was used. All reported p values are two-sided.

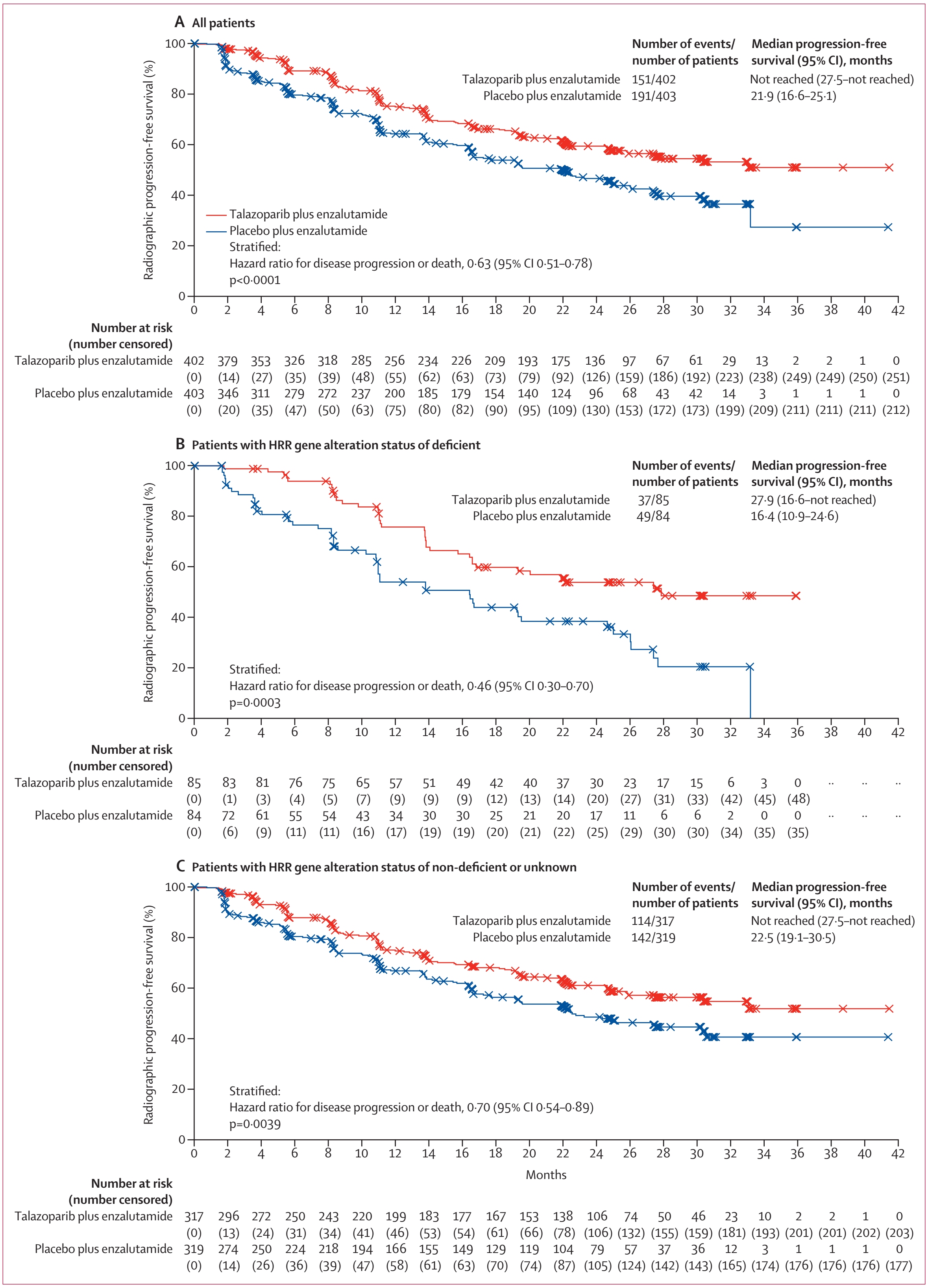
Safety analyses included the safety population, defined as all patients who received at least one dose of study drug. Adverse events were coded to preferred term and system organ class using the Medical Dictionary for Regulatory Activities (MedDRA) and classified by severity using the National Cancer Institute Common Terminology Criteria for Adverse Events version 4.03.

The sponsor, Pfizer, commissioned an independent external data monitoring committee for ongoing

	Talazoparib plus enzalutamide (n=402)	Placebo plus enzalutamide (n=403)
Age, years	71 (66–76)	71 (65–76)
Race		
White	243 (60%)	255 (63%)
Black or African American	11 (3%)	5 (1%)
Asian	127 (32%)	120 (30%)
Multiracial	0	1 (<1%)
Other*	2 (<1%)	1 (<1%)
Not reported	19 (5%)	21 (5%)
Renal impairment†‡		
None or mild	344 (86%)	347 (86%)
Moderate	42 (10%)	41 (10%)
Baseline serum PSA, μg/L	18·2 (6·9–59·4)	16·2 (6·4–53·4)
Baseline circulating tumour cell count, cells per 7·5 mL of blood	1 (0–7)	1 (0–6)
Gleason score†		
<8	117 (29%)	113 (28%)
≥8	281 (70%)	283 (70%)
Disease site		
Bone (including with soft tissue component)	349 (87%)	342 (85%)
Lymph node	147 (37%)	167 (41%)
Visceral (lung)	45 (11%)	61 (15%)
Visceral (liver)	12 (3%)	16 (4%)
Other soft tissue	37 (9%)	33 (8%)
ECOG performance status		
0	259 (64%)	271 (67%)
1	143 (36%)	132 (33%)
Previous taxane-based chemotherapy§	86 (21%)	93 (23%)
Previous treatment with novel hormonal therapy	23 (6%)	27 (7%)
Abiraterone	21 (5%)	25 (6%)
Orteronel	2 (<1%)	2 (<1%)
HRR gene alteration status by randomisation stratification		
Deficient	85 (21%)	84 (21%)
Non-deficient or unknown	317 (79%)	319 (79%)
HRR gene alteration status by prospective tumour tissue testing¶		
Deficient	85 (21%)	82 (20%)
Non-deficient	207 (51%)	219 (54%)
Unknown	110 (27%)	102 (25%)
BRCA1/2 alteration	27 (7%)	32 (8%)

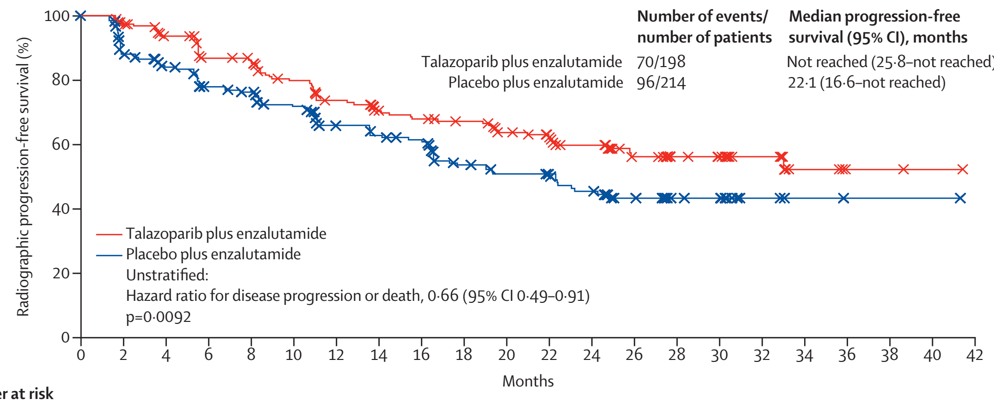
Data are n (%) or median (IQR). ECOG=Eastern Cooperative Oncology Group. HRR=homologous recombination repair. PSA=prostate-specific antigen. *American Indian, Alaska Native, Native Hawaiian, or Other Pacific Islander. †Not reported for the remaining patients. ‡Moderate renal impairment defined as 30–59 mL/min per 1·73 m². §All received docetaxel. ¶Exploratory endpoint analysis by prospective tumour tissue testing to separate unknown from non-deficient HRR gene alteration status. Prospective "unknown" status for HRR gene alteration status primarily reflects the sample(s) submitted to Foundation Medicine not being analysed (typically due to failed quality control metrics) or test results not being reported (due to limitations in sample quality or purity).

Table 1: **Baseline patient demographics and disease characteristics (all-comers intention-to-treat population)**



(Figure 2 continues on next page)

D Patients with HRR gene alteration status of non-deficient by prospective tumour tissue testing*



Number at risk (number censored)	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42
Talazoparib plus enzalutamide	198 (0)	184 (9)	170 (16)	152 (22)	148 (24)	132 (30)	119 (33)	109 (38)	104 (39)	100 (42)	91 (46)	83 (53)	63 (69)	43 (86)	31 (98)	28 (101)	18 (111)	7 (121)	2 (126)	2 (126)	1 (127)	0 (128)
Placebo plus enzalutamide	214 (0)	179 (10)	162 (20)	143 (27)	138 (29)	123 (36)	107 (42)	100 (44)	95 (46)	78 (52)	71 (55)	65 (61)	50 (70)	34 (84)	23 (95)	22 (96)	8 (110)	2 (116)	1 (117)	1 (117)	1 (117)	0 (118)

Figure 2: **Radiographic progression-free survival in all patients and according to HRR gene alteration status (assessed by blinded independent central review; all-comers intention-to-treat population)**
HRR=homologous recombination repair. *Figure 2D shows an exploratory endpoint analysis by prospective tumour tissue testing to separate unknown from non-deficient HRR gene alteration status.

monitoring of efficacy and safety. TALAPRO-2 is registered with ClinicalTrials.gov (NCT03395197).

Role of the funding source

The sponsor (Pfizer) was involved in the trial design (together with the academic steering committee), data analysis, and data interpretation, and funded medical writing support. Astellas Pharma provided enzalutamide. All authors, including those employed by the sponsor, contributed to data interpretation as well as development, writing, and approval of the manuscript.

Results

Between Jan 7, 2019, and Sept 17, 2020, 805 patients receiving ongoing androgen deprivation therapy were enrolled and randomly assigned (402 to the talazoparib group and 403 to the placebo group; intention-to-treat population; figure 1). At least one treatment dose was received by 398 patients in the talazoparib group and 401 patients in the placebo group (all-comers safety population). Important protocol deviations in at least 10% of patients are listed in the appendix (p 16). Data cutoff was Aug 16, 2022. Baseline demographic and disease characteristics were well balanced (table 1; appendix p 17).

Of 805 enrolled patients, all 805 had prospective tumour tissue test results. Of these 805 patients, 115 also had blood-based samples that underwent ctDNA testing after a protocol amendment allowed concurrent prospective ctDNA testing in addition to the tumour tissue testing.

Based on HRR gene alteration status by randomisation stratification, 169 (21%) patients had HRR gene alterations detected (HRR-deficient) and 636 (79%) had no HRR gene alterations detected (HRR-non-deficient) or were of unknown alteration status (table 1). *BRCA* alterations were detected in 27 (7%) patients in the talazoparib group and 32 (8%) in the placebo group. A summary of HRR gene alterations is shown in the appendix (p 18).

Median follow-up for rPFS was 24·9 months (IQR 21·9–30·2) for the talazoparib group and 24·6 months (14·4–30·2) for the placebo group. Treatment with talazoparib plus enzalutamide resulted in a 37% lower risk of radiographic progression (blinded independent central review) or death than placebo plus enzalutamide (HR 0·63; 95% CI 0·51–0·78; p<0·0001; figure 2A). Median rPFS was not reached (95% CI 27·5 months–not reached) in the talazoparib group and was 21·9 months (16·6–25·1) in the placebo group.

In subgroup analysis by HRR gene alteration status, the HR for rPFS was 0·46 (95% CI 0·30–0·70; p=0·0003) in patients with a status of deficient (figure 2B) and 0·70 (0·54–0·89; p=0·0039) in patients with a status of non-deficient or unknown (figure 2C) in favour of talazoparib plus enzalutamide versus placebo plus enzalutamide. Patients with a non-deficient status by prospective tumour tissue testing had a 34% lower risk of radiographic progression or death with talazoparib plus enzalutamide than placebo plus enzalutamide (HR 0·66; 0·49–0·91; p=0·0092; figure 2D, figure 3B). Patients in the talazoparib group whose tumours had *BRCA* gene alterations had a 77% lower risk of radiographic progression or death (HR 0·23; 95% CI 0·10–0·53; p=0·0002) and those with non-*BRCA* HRR gene alterations had a 34% lower risk (HR 0·66; 0·39–1·12; p=0·12) compared with those in the placebo group (figure 3B).

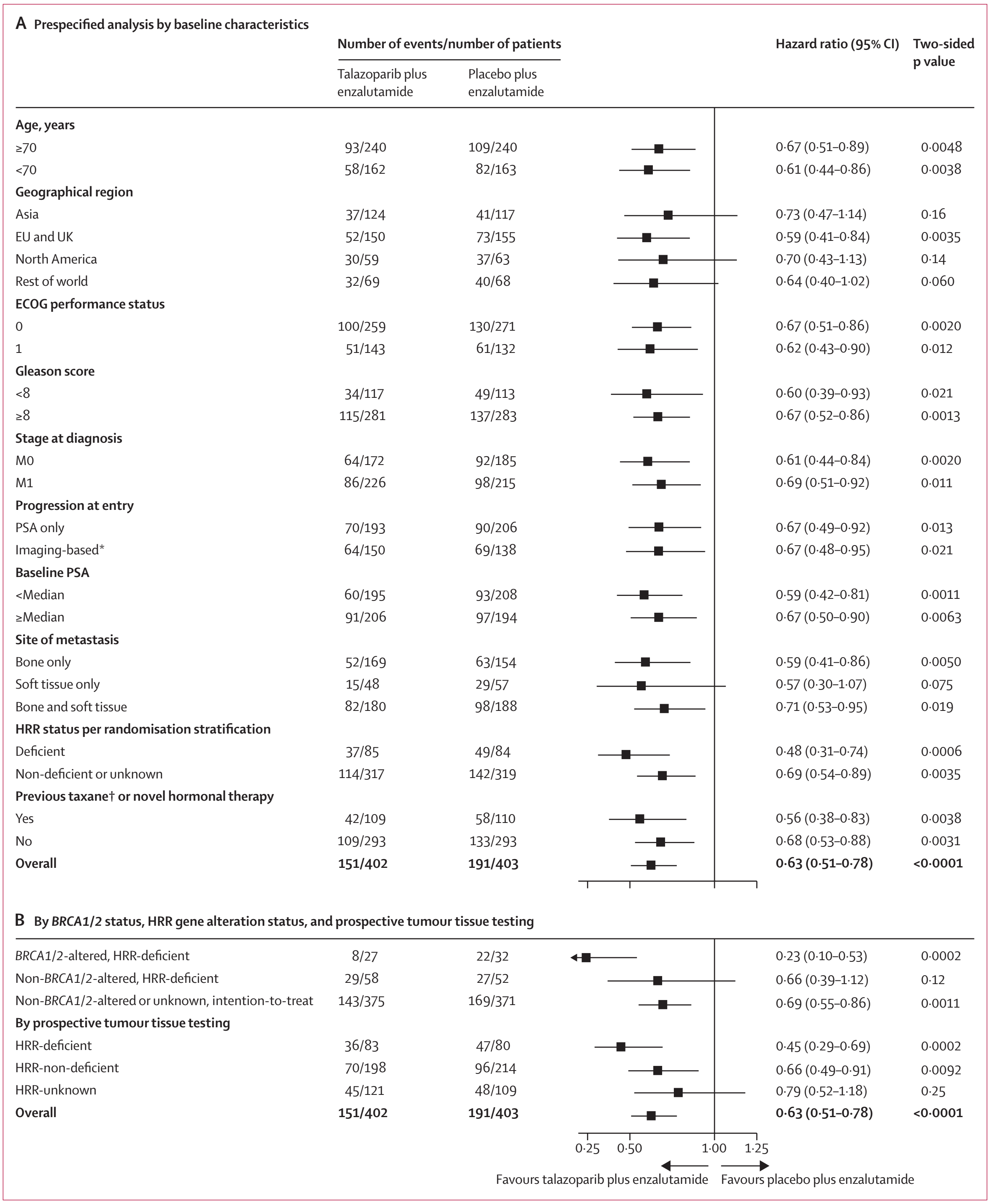


***Figure 3:* Subgroup analysis of radiographic progression-free survival (assessed by blinded independent central review; all-comers intention-to-treat population)**

The overall hazard ratio for all patients, *BRCA1*/2 altered, non-*BRCA1*/2-altered, and non-*BRCA1*/2-altered or unknown, was based on a Cox model stratified by the randomisation stratification factors. For other subgroups, the hazard ratio was based on an unstratified Cox model with treatment as the only covariate. ECOG=Eastern Cooperative Oncology Group. HRR=homologous recombination repair. PSA=prostate-specific antigen. *With or without PSA progression. †All received docetaxel.

A consistent treatment effect for rPFS was observed across prespecified clinical subgroups (figure 3A) and by investigator assessment (appendix p 11). Among 219 patients who had received previous treatment with docetaxel or a novel hormonal therapy for castration-sensitive disease, the HR for rPFS was 0·56 (95% CI

	Talazoparib plus enzalutamide (n=402)	Placebo plus enzalutamide (n=403)	Hazard ratio (95% CI)	p value (two-sided)
Best overall response*, n/N (%)				
Complete response	45/120 (38%)	24/132 (18%)	..	..
Partial response	29/120 (24%)	34/132 (26%)	..	..
Stable disease	36/120 (30%)	38/132 (29%)	..	..
Progressive disease	7/120 (6%)	30/132 (23%)	..	..
Not evaluable	3/120 (2%)	6/132 (5%)	..	..
Objective response*, n/N (%; 95% CI)	74/120 (62%; 52·4–70·4)	58/132 (44%; 35·3–52·8)	..	0·0050
Median duration of response† (95% CI), months	20·4 (16·1–not reached)	19·8 (12·9–not reached)	..	..
PSA response ≥50%‡, n/N (%; 95% CI)	331/396 (84%; 79·6–87·1)	284/394 (72%; 67·4–76·5)	..	0·0001
Time to PSA progression				
Patients with progression, n/N (%)	164/402 (41%)	177/403 (44%)	..	..
Median time to progression (95% CI), months	26·7 (21·2–30·4)	17·5 (14·1–20·8)	0·72 (0·58–0·89)	0·0020
Time to initiation of cytotoxic chemotherapy				
Patients with use, n/N (%)	83/402 (21%)	139/403 (34%)	..	..
Median time to use (95% CI), months	Not reached (37·0–not reached)	Not reached (32·3–not reached)	0·49 (0·38–0·65)	<0·0001
Time to initiation of subsequent antineoplastic therapy				
Patients with use, n/N (%)	114/402 (28%)	176/403 (44%)	..	..
Median time to use (95% CI), months	Not reached (37·0–not reached)	28·3 (23·5–not reached)	0·54 (0.42–0·68)	<0·0001
Time to first symptomatic skeletal event				
Patients with event, n/N (%)	91/402 (23%)	93/403 (23%)	..	..
Median time to first event (95% CI), months	Not reached (not reached)	Not reached (not reached)	0·88 (0·66–1·18)	0·41
PFS2§				
Patients with event, n/N (%)	126/402 (31%)	143/403 (35%)	..	..
Median PFS2 (95% CI), months	36·4 (33·5–not reached)	35·3 (28·6–not reached)	0·77 (0·61–0·98)	0·036

PFS2=progression-free survival 2. PSA=prostate-specific antigen. *Only includes patients with measurable soft tissue disease at baseline per blinded independent central review: talazoparib plus enzalutamide (n=120); placebo plus enzalutamide (n=132). †Only includes patients with confirmed complete response or partial response: talazoparib plus enzalutamide (n=74); placebo plus enzalutamide (n=58). ‡Only includes patients with a baseline PSA value and at least one post-baseline PSA value: talazoparib plus enzalutamide (n=396); placebo plus enzalutamide (n=394). §PFS2 based on investigator assessment (time from randomisation to the date of documented progression on the first subsequent antineoplastic therapy or death from any cause, whichever occurs first).

Table 2: **Secondary efficacy endpoints (all-comers intention-to-treat population)**

0·38–0·83; p=0·0038) in favour of talazoparib plus enzalutamide. Among patients who had received docetaxel (n=179), the HR for rPFS was 0·51 (0·32–0·81; p=0·0034) and among those who had received a novel hormonal agent (n=50), the HR for rPFS was 0·57 (0·28–1·16; p=0·12).

Overall survival data are immature; 123 (31%) of 402 patients in the talazoparib group and 129 (32%) of 403 in the placebo group had died at data cutoff. The HR for death was 0·89 (95% CI 0·69–1·14; p=0·35; appendix p 12) in favour of talazoparib plus enzalutamide. Survival follow-up will continue as planned.

Confirmed objective response rate in patients with measurable disease at baseline by blinded independent central review was 62% (74 of 120; 95% CI 52·4–70·4) for the talazoparib group and 44% (58 of 132; 95% CI 35·3–52·8) for the placebo group, with a complete response in 45 (38%) of 120 and 24 (18%) of 132 patients, respectively (table 2). Treatment with talazoparib plus enzalutamide significantly prolonged the time to PSA progression (HR 0·72; 95% CI 0·58–0·89; p=0·0020), time to initiation of cytotoxic chemotherapy (HR 0·49; 0·38–0·65; p<0·0001), and time to progression or death on first subsequent antineoplastic therapy (HR 0·77; 0·61–0·98; p=0·036) compared with placebo plus enzalutamide (see table 2 for results for other secondary efficacy endpoints). Results for secondary efficacy endpoints by deficient and non-deficient or unknown HRR gene alteration status are in the appendix (pp 19–20). A summary of selected subsequent systemic therapies for prostate cancer is shown in the appendix (p 21).

Median duration of treatment was 19·8 months (IQR 8·8–26·9) for talazoparib and 22·2 months (9·9–28·1) for enzalutamide in the talazoparib group, and 16·1 months (6·5–25·0) for placebo and 16·6 months (6·7–25·1) for enzalutamide in the placebo group.

	Talazoparib plus enzalutamide (n=398)		Placebo plus enzalutamide (n=401)	
	All grades	Grade ≥3	All grades	Grade ≥3
Any adverse event	392 (98%)	299 (75%)	379 (95%)	181 (45%)
Treatment-related adverse event	357 (90%)	234 (59%)	279 (70%)	71 (18%)
Serious adverse event	157 (39%)	145 (36%)	107 (27%)	94 (23%)
Serious and treatment-related adverse event	78 (20%)	68 (17%)	12 (3%)	11 (3%)
Adverse event resulting in dose interruption of:				
Talazoparib or placebo*	247 (62%)	..	84 (21%)	..
Enzalutamide†	156 (39%)	..	78 (19%)	..
Adverse event resulting in dose reduction of:				
Talazoparib or placebo*	210 (53%)	..	27 (7%)	..
Enzalutamide†	58 (15%)	..	32 (8%)	..
Adverse event resulting in permanent drug discontinuation of:				
Talazoparib or placebo*	75 (19%)	..	49 (12%)	..
Enzalutamide†	43 (11%)	..	44 (11%)	..
Grade 5 adverse event	13 (3%)‡	..	18 (4%)§	..
Most common adverse events (all grades in ≥10% of patients)¶				
Anaemia	262 (66%)	185 (46%)	70 (17%)	17 (4%)
Neutropenia	142 (36%)	73 (18%)	28 (7%)	6 (1%)
Fatigue	134 (34%)	16 (4%)	118 (29%)	8 (2%)
Thrombocytopenia	98 (25%)	29 (7%)	14 (3%)	4 (1%)
Back pain	88 (22%)	10 (3%)	72 (18%)	4 (1%)
Leukopenia	88 (22%)	25 (6%)	18 (4%)	0
Decreased appetite	86 (22%)	5 (1%)	63 (16%)	4 (1%)
Nausea	82 (21%)	2 (<1%)	50 (12%)	3 (<1%)
Constipation	72 (18%)	1 (<1%)	68 (17%)	2 (<1%)
Fall	71 (18%)	9 (2%)	59 (15%)	8 (2%)
Arthralgia	58 (15%)	2 (<1%)	79 (20%)	2 (<1%)
Asthenia	57 (14%)	11 (3%)	38 (9%)	3 (<1%)
Diarrhoea	57 (14%)	1 (<1%)	55 (14%)	0
Hypertension	55 (14%)	21 (5%)	62 (15%)	30 (7%)
Dizziness	48 (12%)	4 (1%)	23 (6%)‖	2 (<1%)
Hot flush	47 (12%)	0	53 (13%)‖	0
Lymphopenia	45 (11%)	20 (5%)	20 (5%)	4 (1%)
Oedema peripheral	42 (11%)	0	23 (6%)‖	0
Dyspnoea	41 (10%)	2 (<1%)	25 (6%)	1 (<1%)
Decreased weight	40 (10%)	2 (<1%)	33 (8%)	3 (<1%)

Data are n (%). Shown are adverse events that occurred from the time of the first dose of study treatment through to 28 days after permanent discontinuation of all study treatments or before initiation of a new antineoplastic or any investigational therapy, whichever occurred first. Adverse events were graded according to National Cancer Institute Common Terminology Criteria for Adverse Events version 4.03. *Includes permanent discontinuation, dose reduction, or dose interruption of talazoparib or placebo only plus permanent discontinuation, dose reduction, or dose interruption of both talazoparib or placebo and enzalutamide. †Includes permanent discontinuation, dose reduction, or dose interruption of enzalutamide only plus permanent discontinuation, dose reduction, or dose interruption of both talazoparib or placebo and enzalutamide. ‡None were considered treatment-related. §Two were considered treatment-related. ¶None of these events were recorded as grade 5. ‖One additional patient in each of these categories had a missing or unknown grade.

Table 3: **Summary of adverse events (all-comers safety population)**

Median relative dose intensities in the talazoparib group were 83·5% (IQR 66·2–100·0) for talazoparib and 99·9% (95·6–100·0) for enzalutamide.

The most common all-cause adverse events in the talazoparib group (≥30% of patients) were anaemia, neutropenia, and fatigue (table 3; by grade in appendix p 22). The most common grade 3–4 adverse events (≥10% of patients) were anaemia (185 [46%] of 398) and neutropenia (73 [18%] of 398) in the talazoparib group. To ensure optimal dosing of talazoparib at the individual level, the protocol did not require a dose modification of talazoparib or placebo until the anaemia was grade 3 or higher. At baseline, 195 (49%) of 398 patients in the talazoparib group had grade 1–2 anaemia. During the protocol treatment, 185 (46%) patients in the talazoparib group developed grade 3–4 anaemia after a median of 3·3 months, following which they underwent protocol-mandated dose reduction of talazoparib. Only 33 (8%) patients in the talazoparib group discontinued talazoparib due to anaemia.

There were more dose interruptions and reductions due to adverse events in the talazoparib group than in the placebo group (table 3). The dose of talazoparib was interrupted due to adverse events in 247 (62%) of 398 patients compared with 84 (21%) of 401 patients who had a dose interruption of placebo (the investigators were masked to treatment so were not aware of the talazoparib *vs* placebo assignments in their respective patients). The talazoparib dose was reduced due to adverse events in 210 (53%) of 398 patients and the placebo dose was reduced due to adverse events in 27 (7%) of 401 patients. The most common adverse events leading to a dose reduction of talazoparib were anaemia (172 [43%] of 398 patients), neutropenia (60 [15%]), thrombocytopenia (22 [6%]), and leukopenia (nine [2%]). 75 (19%) of 398 patients discontinued talazoparib and 49 (12%) of 401 patients discontinued placebo due to adverse events. The most common adverse events leading to discontinuation of talazoparib were anaemia (33 [8%] of 398) and neutropenia (13 [3%] of 398). Discontinuation rates of enzalutamide due to adverse events were 11% (43 of 398) in the talazoparib group versus 11% (44 of 401) in the placebo group.

In the talazoparib group, there was one instance (<1%) of myelodysplastic syndrome during the safety reporting period and one instance (<1%) of acute myeloid leukaemia during the follow-up period (none in the placebo group). Venous embolic and thrombotic events were reported in 16 (4%) of 398 patients in the talazoparib group, including ten (3%) patients with pulmonary embolism (grade 3 in nine patients), and in three (<1%) of 401 patients in the placebo group, all of which were grade 3 pulmonary embolisms. Grade 3–4 nausea, decreased appetite, and fatigue were uncommon (<1%–4%) in patients in the talazoparib group. Treatment-related deaths occurred in no patients in the talazoparib group and two patients (<1%) in the placebo group.

Interactive visualisation of the data presented in this article is available at the TALAPRO-2 Dashboard.

Discussion

Across the all-comers population and prespecified subgroups, talazoparib plus enzalutamide resulted in clinically meaningful and statistically significant benefits over enzalutamide plus placebo, which performed in line with extended follow-up of rPFS with enzalutamide alone in the PREVAIL trial (median 20·0 months [95% CI 18·9–22·1]).[21] Statistically significant and clinically meaningful improvements in rPFS were seen for the all-comers population (the primary endpoint), HRR-deficient subgroup, and HRR-non-deficient or unknown subgroup, as well as in the HRR-non-deficient subgroup by prospective tumour tissue testing only. In addition, key secondary endpoints, including time to PSA progression, time to cytotoxic chemotherapy, and time to progression or death on first subsequent antineoplastic therapy, all favoured the talazoparib group. Overall survival data are immature. Further study is ongoing to agnostically identify and explore alterations in single genes or gene pairs (entire FoundationOne panel), and more complex molecular signatures, that might be prognostic or predictive in the HRR-non-deficient population.

The proportion of patients with HRR gene alterations in the enrolled population (talazoparib group, 21%; placebo group, 21%) was in keeping with previous studies,[3,22–24] and *BRCA* alterations were not over-represented (talazoparib group, 7%; placebo group, 8%). Superiority of the talazoparib group was maintained when patients with *BRCA* alterations were excluded from the rPFS efficacy analysis. Patients who received docetaxel or abiraterone for castration-sensitive prostate cancer (around 27% of the population) also had a significant benefit, with a 44% reduction in risk of radiographic progression or death. This is important given that patients with metastatic castration-sensitive prostate cancer increasingly receive intensified treatment of androgen deprivation combined with novel hormonal therapy or chemotherapy, or both. However, only around 6% of patients in either group had been previously treated with abiraterone; the benefit in these patients is hypothesis-generating and warrants further studies.

The TALAPRO-2 results are supported by those from the recent phase 3 PROpel trial (NCT03732820), which demonstrated improved progression-free survival with olaparib plus abiraterone over placebo plus abiraterone in patients with mCRPC.[25] In addition to combining the PARP inhibitor talazoparib with the androgen receptor pathway inhibitor enzalutamide, a key strength of TALAPRO-2 is that biomarker status was obtained prospectively before randomisation and used as a stratification factor. Importantly, in TALAPRO-2, talazoparib plus enzalutamide significantly improved rPFS in patients without HRR gene alterations. In contrast to TALAPRO-2, the phase 3 MAGNITUDE trial (NCT03748641) showed no benefit of adding niraparib to abiraterone in patients without HRR gene alterations detected.[26] The reasons for this differing result are yet to be elucidated but might be related to the trial design or differences in the dosing or characteristics of the individual PARP inhibitors—eg, their ability to trap PARP on DNA.[27] Testing for HRR gene alterations will continue to help guide treatment decisions and allow informed discussions of the risk–benefit ratio for individual patients with mCRPC depending on HRR gene alterations detected.

For the **TALAPRO-2 Dashboard** see https://clinical-trials.dimensions.ai/talapro2/

The safety profile of talazoparib plus enzalutamide in TALAPRO-2 is consistent with the known safety profiles of the individual medicines; however, the grades and frequencies of haematological adverse events were higher than with talazoparib monotherapy.[10,28,29] Treatment-emergent adverse events were generally managed through close monitoring of patients, dose modifications, and supportive measures, including transfusions. For example, to optimise the dosing of talazoparib in the context of the eligibility criterion for haemoglobin of at least 9 g/dL, the protocol did not require dose reduction or interruption of talazoparib or placebo until anaemia was grade 3 or higher (haemoglobin <8 g/dL). Notably, almost half of the patients (49%) had grade 1–2 anaemia at baseline, and 46% developed grade 3–4 anaemia after a median talazoparib treatment duration of 3·3 months. After protocol-mandated dose reduction of talazoparib, only 8% of patients discontinued talazoparib because of anaemia. These data suggest the requirement for close monitoring of blood counts in the first 3–4 months of talazoparib treatment, following which an appropriate dose of talazoparib at the individual level can be established for subsequent long-term treatment. Despite the frequency of dose modifications and the 19% overall incidence of permanent discontinuation of talazoparib due to adverse events, the relative dose intensity in the talazoparib group remained high at more than 80%.

It could be noted as a limitation of the study that the treatment landscape of prostate cancer is evolving rapidly, with many patients now receiving novel hormonal therapy in the castration-sensitive setting. Additionally, and to reflect a more real-world patient population of mCRPC, patients were eligible to enrol in the TALAPRO-2 study with haemoglobin levels of 9 g/dL or higher. As the protocol did not mandate dose modification until haemoglobin levels were less than 8 g/dL to optimise individual treatment, this could have contributed to the high rate of grade 3–4 anaemia that was observed.

In conclusion, results from the primary analysis of the all-comers population of the TALAPRO-2 trial support the consideration of talazoparib plus enzalutamide as a first-line treatment option in patients with mCRPC. Final overall survival data and additional long-term safety follow-up will further clarify the clinical benefit of the treatment combination in patients with and without tumour HRR gene alterations.

Contributors

All authors, including those employed by the sponsor, contributed to data interpretation, and development, writing, and approval of the final manuscript. NA and XL accessed and verified the data. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

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Data sharing

Upon request, and subject to review, Pfizer will provide the data that support the findings of this study. Subject to certain criteria, conditions and exceptions, Pfizer may also provide access to the related individual de-identified participant data. See https://www.pfizer.com/science/clinical-trials/trial-data-and-results for more information. Interactive visualisation of the data presented in this article is available at the TALAPRO-2 Dashboard (https://clinical-trials.dimensions.ai/talapro2/).

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References

1 Sung H, Ferlay J, Siegel RL, et al. Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin* 2021; **71:** 209–49.

2 Siegel RL, Miller KD, Wagle NS, Jemal A. Cancer statistics, 2023. *CA Cancer J Clin* 2023; **73:** 17–48.

3 Chung JH, Dewal N, Sokol E, et al. Prospective comprehensive genomic profiling of primary and metastatic prostate tumors. *JCO Precis Oncol* 2019; **3:** PO.18.00283.

4 Shen Y, Rehman FL, Feng Y, et al. BMN 673, a novel and highly potent PARP1/2 inhibitor for the treatment of human cancers with DNA repair deficiency. *Clin Cancer Res* 2013; **19:** 5003–15.

5 Farmer H, McCabe N, Lord CJ, et al. Targeting the DNA repair defect in *BRCA* mutant cells as a therapeutic strategy. *Nature* 2005; **434:** 917–21.

6 Bryant HE, Schultz N, Thomas HD, et al. Specific killing of BRCA2-deficient tumours with inhibitors of poly(ADP-ribose) polymerase. *Nature* 2005; **434:** 913–17.

7 Lord CJ, Ashworth A. PARP inhibitors: synthetic lethality in the clinic. *Science* 2017; **355:** 1152–58.

8 Tran C, Ouk S, Clegg NJ, et al. Development of a second-generation antiandrogen for treatment of advanced prostate cancer. *Science* 2009; **324:** 787–90.

9 Murai J, Huang SY, Renaud A, et al. Stereospecific PARP trapping by BMN 673 and comparison with olaparib and rucaparib. *Mol Cancer Ther* 2014; **13:** 433–43.

10 de Bono JS, Mehra N, Scagliotti GV, et al. Talazoparib monotherapy in metastatic castration-resistant prostate cancer with DNA repair alterations (TALAPRO-1): an open-label, phase 2 trial. *Lancet Oncol* 2021; **22:** 1250–64.

11 de Bono J, Mateo J, Fizazi K, et al. Olaparib for metastatic castration-resistant prostate cancer. *N Engl J Med* 2020; **382:** 2091–102.

12 Fizazi K, Piulats JM, Reaume MN, et al. Rucaparib or physician's choice in metastatic prostate cancer. *N Engl J Med* 2023; **388:** 719–32.

13 Rao A, Moka N, Hamstra DA, Ryan CJ. Co-inhibition of androgen receptor and PARP as a novel treatment paradigm in prostate cancer—where are we now? *Cancers (Basel)* 2022; **14:** 801.

14 Asim M, Tarish F, Zecchini HI, et al. Synthetic lethality between androgen receptor signalling and the PARP pathway in prostate cancer. *Nat Commun* 2017; **8:** 374.

15 Polkinghorn WR, Parker JS, Lee MX, et al. Androgen receptor signaling regulates DNA repair in prostate cancers. *Cancer Discov* 2013; **3:** 1245–53.

16 Gui B, Gui F, Takai T, et al. Selective targeting of PARP-2 inhibits androgen receptor signaling and prostate cancer growth through disruption of FOXA1 function. *Proc Natl Acad Sci USA* 2019; **116:** 14573–82.

17 Schiewer MJ, Goodwin JF, Han S, et al. Dual roles of PARP-1 promote cancer growth and progression. *Cancer Discov* 2012; **2:** 1134–49.

18 Agarwal N, Azad A, Shore ND, et al. Talazoparib plus enzalutamide in metastatic castration-resistant prostate cancer: TALAPRO-2 phase III study design. *Future Oncol* 2022; **18:** 425–36.

19 Agarwal N, Shore ND, Dunshee C, et al. Clinical and safety outcomes of TALAPRO-2: a two-part phase III study of talazoparib (TALA) in combination with enzalutamide (ENZA) in metastatic castration-resistant prostate cancer (mCRPC). *J Clin Oncol* 2019; **37** (suppl)**:** 5076.

20 US Department of Health and Human Services. Common terminology criteria for adverse events (CTCAE) version 4.03. https://evs.nci.nih.gov/ftp1/CTCAE/CTCAE_4.03/CTCAE_4.03_2010-06-14_QuickReference_8.5x11.pdf (accessed April 1, 2023).

21 Beer TM, Armstrong AJ, Rathkopf D, et al. Enzalutamide in men with chemotherapy-naïve metastatic castration-resistant prostate cancer: extended analysis of the phase 3 PREVAIL study. *Eur Urol* 2017; **71:** 151–54.

22 Abida W, Armenia J, Gopalan A, et al. Prospective genomic profiling of prostate cancer across disease states reveals germline and somatic alterations that may affect clinical decision making. *JCO Precis Oncol* 2017; **1:** PO.17.00029.

23 Armenia J, Wankowicz SAM, Liu D, et al. The long tail of oncogenic drivers in prostate cancer. *Nat Genet* 2018; **50:** 645–51.

24 Robinson D, Van Allen EM, Wu YM, et al. Integrative clinical genomics of advanced prostate cancer. *Cell* 2015; **161:** 1215–28.

25 Clarke NW, Armstrong AJ, Thiery-Vuillemin A, et al. Abiraterone and olaparib for metastatic castration-resistant prostate cancer. *NEJM Evid* 2022; **1:** EVIDoa2200043.

26 Chi KN, Rathkopf D, Smith MR, et al. Niraparib and abiraterone acetate for metastatic castration-resistant prostate cancer. *J Clin Oncol* 2023; **41:** 3339–51.

27 Zandarashvili L, Langelier MF, Velagapudi UK, et al. Structural basis for allosteric PARP-1 retention on DNA breaks. *Science* 2020; **368:** eaax6367.

28 Beer TM, Armstrong AJ, Rathkopf DE, et al. Enzalutamide in metastatic prostate cancer before chemotherapy. *N Engl J Med* 2014; **371:** 424–33.

29 Mehra N, Fizazi K, de Bono JS, et al. Talazoparib, a poly (ADP-ribose) polymerase inhibitor, for metastatic castration-resistant prostate cancer and DNA damage response alterations: TALAPRO-1 safety analyses. *Oncologist* 2022; **27:** e783–95.